SF2943 Project — Part A

Daily electricity load for Swedish bidding zone SE_3

1 Data

We model daily mean electricity load (MW) for the Swedish bidding zone **SE_3** (Stockholm, Mälardalen, eastern Sweden), sourced from *Open Power System Data* (OPSD), `time_series_60min_singleindex.csv`, column `SE_3_load_actual_entsoe_transparency` (originally ENTSO-E Transparency Platform).[1] The hourly file covers 2015-01-01 01:00 to 2020-09-30 22:00 UTC (50 398 hourly observations, 77 NaN, longest gap 49 h). We aggregate to daily means (units preserved); days with fewer than 20 valid hours are marked NaN and the remaining 5 NaN days are linearly interpolated, giving $n = 2100$ daily observations X_t, well above the $N \geq 2000$ threshold.

2 Cleaning to stationarity

Raw X_t has a winter peak / summer trough cycle, a weekday/weekend cycle, a mild secular trend and an amplitude that grows with the level. We use the classical decomposition (B&D §1.5) on the log-transformed series:

$$\log X_t \;=\; m_t \;+\; s_t^{(\text{year})} \;+\; s_t^{(\text{week})} \;+\; Y_t, \tag{1}$$

fitted in the order *year* → *week* → *trend* by ordinary least squares (OLS), each stage operating on the residual of the previous one. The Box–Cox transform with $\lambda = 0$ (i.e. log) stabilises variance and turns multiplicative seasonality into additive, justifying (1).

Yearly seasonality is a $k = 2$ harmonic (B&D §1.3) with period $d = 365$:

$$\hat{s}_t^{(\text{year})} \;=\; \sum_{j=1}^{2} \big[a_j \cos(2\pi j t/365) + b_j \sin(2\pi j t/365)\big],$$

with $(a_1, b_1, a_2, b_2) = (+0.2384, +0.0773, -0.0127, -0.0013)$. The fundamental captures the dominant winter–summer swing; the second harmonic absorbs the asymmetric peak shape.

Weekly seasonality is a $k = 3$ harmonic with period $d = 7$ fitted to the year-deseasonalised series; its coefficients are $(a_1, \ldots, b_3) = (+0.0323, -0.0536, +0.0096, +0.0304, -0.0086, -0.0107)$. Since $d = 7$ is odd, $k = 3$ is the saturated Fourier basis for the weekly cycle and is mathematically equivalent to a day-of-week dummy regression.

Polynomial trend is fitted on the deseasonalised series. We start at degree 1; if the residual mean of any third of the series exceeds 10% of the residual standard deviation we promote to degree 2. Here the linear residual thirds were $(-0.0061, +0.0113, -0.0052)$ with std 0.0645 (max ratio ≈ 0.18), so we adopt the quadratic

$$\hat{m}_t \;=\; 9.1609 \;+\; (6.12 \times 10^{-5})\, t \;-\; (3.35 \times 10^{-8})\, t^2.$$

The cleaning residual $\hat{Y}_t = \log X_t - \hat{m}_t - \hat{s}_t^{(\text{year})} - \hat{s}_t^{(\text{week})}$ is shown in Figure 1c; visually it has constant level and amplitude.

Diagnostics. As expected the iid null is rejected (Ljung–Box $Q(20) = 4134$, $p \approx 0$): cleaning removes the deterministic structure but leaves stationary stochastic dependence for the ARMA stage. The sample ACF (Figure 2a) decays slowly with $\hat{\rho}(1) = 0.83$, has a clear lag-7 spike $\hat{\rho}(7) = 0.30$, and a small residual yearly signal $\hat{\rho}(365) = 0.16$. The PACF (Figure 2b) shows its largest spike at lag 1 with smaller activity at lags 2–3 — characteristic of an AR-dominated process. As a sanity check, $\nabla_1 \nabla_7 \log X_t$ (B&D §1.5.2) gives a qualitatively similar ACF.

3 ARMA fitting

Driven by the ACF/PACF shape we fit five candidates by Gaussian MLE (B&D §5.2; the innovations algorithm gives the conditional likelihood and `statsmodels.tsa.arima.model.ARIMA` maximises it numerically). All five fits are causal and invertible: every root of $\hat{\phi}(z)$ and $\hat{\theta}(z)$ lies outside the unit circle. Order selection is by AICC (B&D §5.5.2), AICC = AIC$+2k(k+1)/(n-k-1)$ with $k = p+q+2$ counting the constant and σ^2. AICC values (params $= p+q$): ARMA(2,1) -8083.73 ✓, ARMA(3,0)

[1] `https://data.open-power-system-data.org/time_series/`

-8083.68, ARMA(1,1) -8081.53 ✓, ARMA(2,0) -8075.06, ARMA(1,0) -8036.14. ARMA(2,1) and AR(3) are essentially tied (ΔAICC $= 0.05$, well below the conventional indistinguishability threshold of 2). We adopt **ARMA(2,1)** as primary and use AR(3) as a Yule–Walker cross-check.

Chosen model. ARMA(2,1) MLE: $\hat{\phi}_1 = +0.441$ (0.094), $\hat{\phi}_2 = +0.279$ (0.082), $\hat{\theta}_1 = +0.512$ (0.091), $\hat{\sigma}^2 = 1.24\times10^{-3}$; constant -0.0004 (0.004) is insignificant ($p = 0.93$) as expected since $\hat{Y}_t$ is mean-zero by OLS construction. AR roots $z \in \{1.26, -2.84\}$, MA root -1.95, all outside the unit disc. **Yule–Walker** (B&D §5.1.1) on the AR(3) candidate gives $\hat{\phi}^{\mathrm{YW}} = (+0.948, -0.206, +0.075)$, $\hat{\sigma}^2 = 1.25\times10^{-3}$, matching the AR(3) MLE $(+0.954, -0.207, +0.071)$ to three decimals — two different estimators target the same population quantity and agree. **Residual diagnostics.** Ljung–Box on ARMA(2,1) residuals with the B&D §5.3 correction df $= h - p - q$ gives $Q(20) = 43.7$, df $= 17$, $p = 4\times10^{-4}$ — the model fails residual whiteness. The surviving correlation is concentrated at multiples of 7 (Figure 2c): the harmonic weekly basis captures the deterministic weekday *shape* but leaves shared noise structure between same-weekdays that a low-order ARMA cannot absorb. A SARIMA component at lag 7 would target it but is outside the course toolkit; we report the failure rather than overparameterise.

4 Forecasting

We hold out the last 30 days (2020-09-01 to 2020-09-30) and refit ARMA(2,1) on the first $n - 30$ residuals. The h-step-ahead forecast and its prediction MSE $v_n(h)$ are computed via the ARMA recursion (B&D §3.3.1; `statsmodels.get_forecast` runs the innovations algorithm internally). The Gaussian 95% PI $\hat{Y}_{n+h} \pm 1.96\sqrt{v_n(h)}$ is inverted to MW by adding the extrapolated $\hat{m}_t + \hat{s}_t^{(\mathrm{year})} + \hat{s}_t^{(\mathrm{week})}$ and exponentiating; the MW interval is therefore asymmetric (log-normal) but its 95% probabilistic coverage is preserved by monotonicity. Exponentiating the conditional mean yields the conditional *median* of X_{n+h} (the mean differs by $\exp(v_n(h)/2) \leq 1.0025$, negligible here).

Skill (held-out 30 days). RMSE $= 187.6$ MW, MAE $= 145.1$ MW, RMSE/std(train) $= 0.097$. Empirical 95% PI coverage is 30/30, statistically consistent with the nominal level ($0.95^{30} \approx 0.21$). Forecast and actual agree to within ~ 250 MW on every test day (Figure 3); beyond a few days the residual forecast decays to zero and the medium-horizon forecast is dominated by the deterministic seasonal/trend extrapolation.

5 Reflection

Difficulties. The most stubborn feature of $\hat{Y}_t$ is the lag-7 stochastic dependence: a harmonic weekly basis cannot represent it, and no low-order ARMA fully absorbs it within the course toolkit (no SARIMA). We report the Ljung–Box failure rather than overparameterise. **Robustness.** Forecasts are insensitive to model order within the candidate set: ARMA(2,1) and AR(3) give point forecasts that differ by ≤ 4 MW on every test day, and PI widths by $\leq 1\%$. The dominant forecast error is the deterministic component: any holiday, weather event or secular shock not in the seasonal model is mispredicted. The 30-day window is one realisation; a rolling-origin evaluation would tighten the skill estimate. We note one mild data leak — the deterministic components in the test window are evaluated from the full-sample OLS fit, with practical effect on the order of 1% since OLS coefficients depend only weakly on any 30 of 2100 observations. **Alternatives.** (i) SARIMA$(p,0,q)(P,0,Q)_7$ would absorb the lag-7 carryover. (ii) Exogenous regressors (heating-degree-days, holiday indicators, COVID-19 mobility) would explain the largest individual-day errors. (iii) Period 365.25 would remove the phase drift visible in $\hat{\rho}(365) = 0.16$. (iv) Joint OLS for trend+seasonals (vs sequential) would mildly reduce coefficient variance. **Software.** Python 3 with `statsmodels` 0.14 (`ARIMA` for Gaussian MLE; `yule_walker` for the moment estimator; `acorr_ljungbox` with `model_df` for the corrected portmanteau test; `plot_acf`, `plot_pacf` for diagnostics), `numpy` (`linalg.lstsq` for OLS), `pandas`, `matplotlib`. Reproduce the pipeline with `jupyter nbconvert --execute project_part_a.ipynb` and the report figures with `make_report_figures.py`.

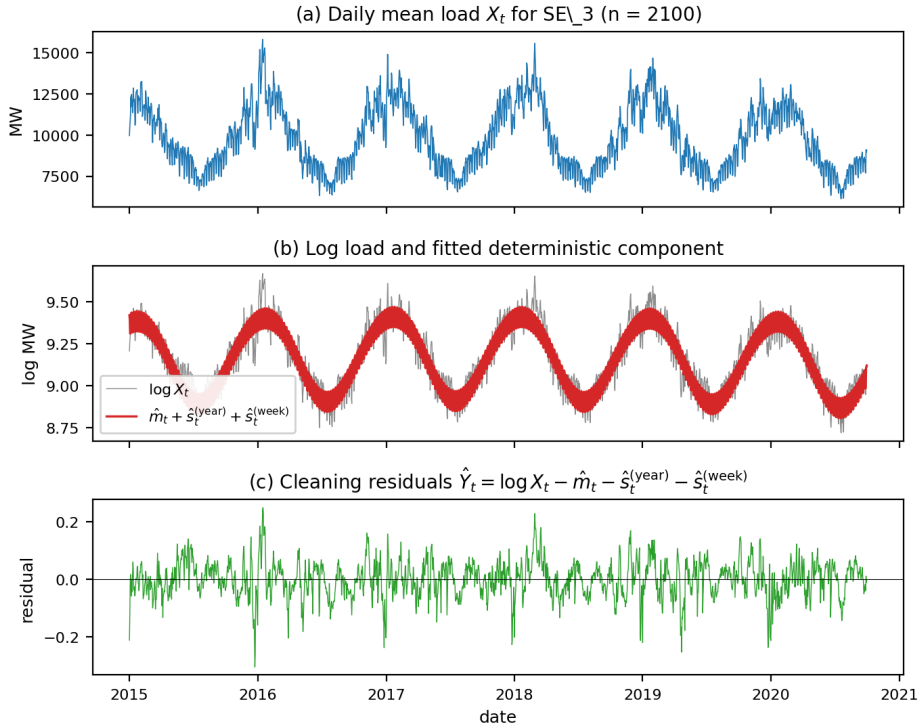

Figure 1: Classical decomposition (B&D §1.5) of daily SE_3 load, 2015-01-01 to 2020-09-30. (a) Raw daily mean load X_t, $n = 2100$. (b) Log-load $\log X_t$ (grey) overlaid with the fitted deterministic component $\hat{m}_t + \hat{s}_t^{(\text{year})} + \hat{s}_t^{(\text{week})}$ (red); seasonals are harmonic regressions ($d = 365, k = 2$ and $d = 7, k = 3$) and the trend is a quadratic OLS fit. (c) Cleaning residuals $\hat{Y}_t$ entering the ARMA stage.

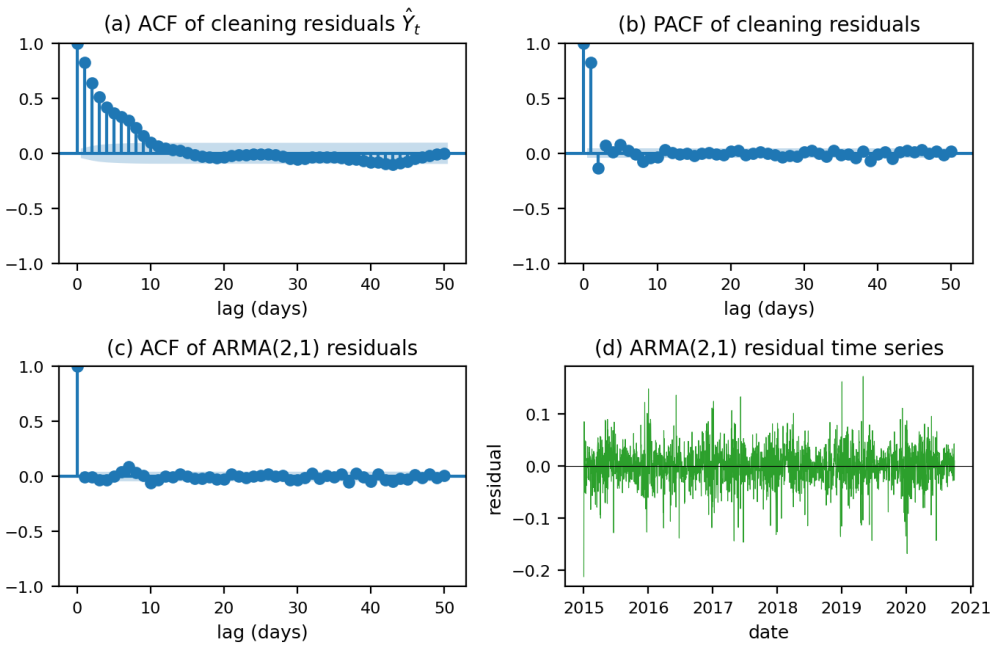

Figure 2: Diagnostic plots. (a) Sample ACF of cleaning residuals $\hat{Y}_t$ (lags 0–50, blue bands at $\pm 1.96/\sqrt{n}$, B&D §1.4.1). Slow decay $\hat{\rho}(1) = 0.83$ and the lag-7 spike $\hat{\rho}(7) = 0.30$ motivate an AR-dominated ARMA fit. (b) Sample PACF of $\hat{Y}_t$, largest spike at lag 1. (c) ACF of ARMA(2,1) residuals; the surviving correlation is concentrated at multiples of 7 ($Q(20) = 43.7$, df $= 17$, $p = 4 \times 10^{-4}$). (d) ARMA(2,1) residual time series; level and amplitude stable.

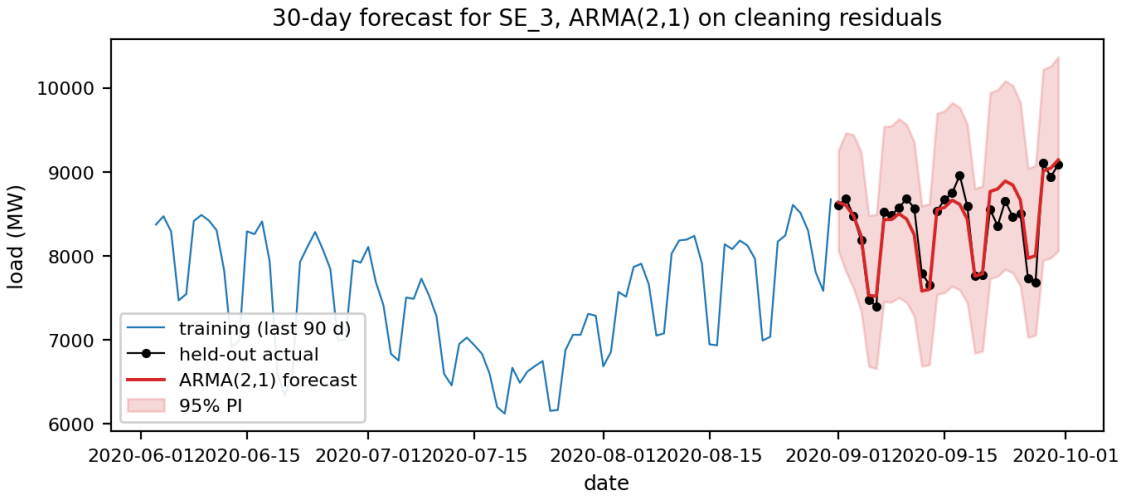

Figure 3: 30-day forecast on the held-out window 2020-09-01 to 2020-09-30, ARMA(2,1) refit on the first $n - 30$ cleaning residuals. Blue: last 90 days of training. Black markers: held-out actual daily means. Red: h-step ARMA forecast (B&D §3.3.1) inverted to MW by adding the extrapolated trend and seasonals and exponentiating; red band is the 95% PI (log-normal in MW). RMSE = 187.6 MW, MAE = 145.1 MW, empirical PI coverage 30/30.